Modeling Assignment 9: Poisson and ZIP Regression Models

Assignment Overview

In this assignment we will be fitting models and calculating the various summative statistics that are associated with Poisson and Zero-Inflated Poisson Regression. The data set for this assignment, STRESS, includes information from about 650 adolescents in the United States who were surveyed about the number of stressful life events they had experienced in the past year (STRESS). STRESS is also an integer variable that represents counts of stressful events. The dataset also includes school and family related variables, which are assumed to be continuously distributed. The variables in this data set are:

COHES = measure of how well the adolescent gets along with their family (coded low to high)
ESTEEM = measure of self-esteem (coded low to high)
GRADES = past year's school grades (coded low to high)
SATTACH = measure of how well the adolescent likes and is attached to their school (coded low to high)

There is no other information about this data or the variables.

Assignment Tasks

1. For the STRESS variable, make a histogram and obtain summary statistics. Obtain a normal probability (Q-Q) plot for the STRESS variable. Is STRESS a normally distributed variable? What do you think is its most likely probability distribution for STRESS? Give a justification for the distribution you selected.

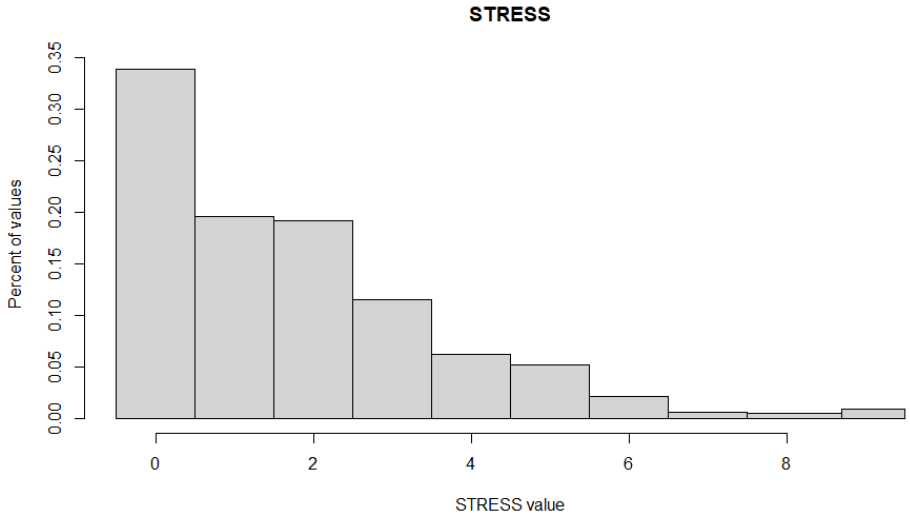

Figure 1. Histogram of STRESS variable with counts on the left, maxing out at 221 observations of 0.

a. This is a large proportion of the values (about 1/3), with all positive counts, so we will likely look to use Zero-Inflated Poisson (ZIP) or Zero-Inflated Negative Binomial (ZINB). Based on the tail and continued decrease from 1 to 2 and so on, my guess would be either NB or ZINB. Oftentimes NB will be useful when the data is overdispersed, so we can check the mean vs variance. We find a mean of 1.730 and variance of 3.419, which is close enough for now to continue with Poisson, though NB could be tested as well to see if it better fits the data. STRESS is not normally distributed as there is a hard cutoff at 0 and no typical "bell curve".
b. Our summary stats are as follows for the STRESS variable; min: 0, 1st Q: 0, median: 1, mean: 1.73, 3rd Q: 3, and max: 9.

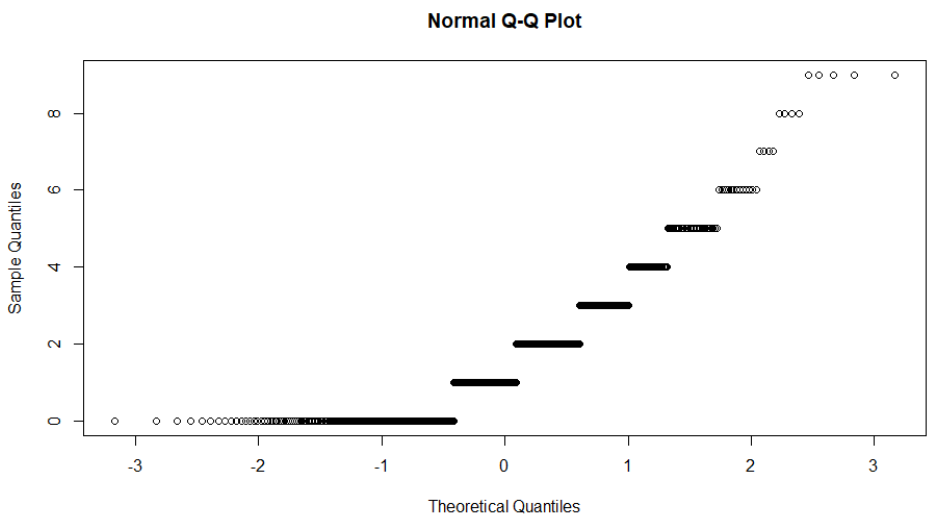

Figure 2. Normal Q-Q plot for the STRESS variable. This shows a few potential outliers at the top but many more observations than we would expect at a negative deviance greater than 2, likely due to the large number of 0s in the data.

2. Fit an OLS regression model to predict STRESS (Y) using COHES, ESTEEM, GRADES, SATTACH as explanatory variables (X). Obtain the typical diagnostic information and graphs. Discuss how well this model fits. Obtain predicted values (Y_hat) and plot them in a histogram. What issues do you see?

```
> summary(ols)

Call:
lm(formula = STRESS ~ COHES + ESTEEM + GRADES + SATTACH, data = md)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.1447 -1.3827 -0.3819  0.9504  6.9525 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  5.71281    0.58118   9.830  < 2e-16 ***
COHES       -0.02319    0.00703  -3.298  0.00103 ** 
ESTEEM      -0.04129    0.01933  -2.136  0.03305 *  
GRADES      -0.04170    0.02352  -1.773  0.07670 .  
SATTACH     -0.03042    0.01412  -2.154  0.03160 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.776 on 646 degrees of freedom
Multiple R-squared:  0.08319,	Adjusted R-squared:  0.07751 
F-statistic: 14.65 on 4 and 646 DF,  p-value: 1.826e-11
```

Output 1. Summary output for lm() call for STRESS predicted by the explanatory variables present. While all variables are significant to the 0.1 threshold, GRADES is not significant at the typical 0.05 threshold. Additionally, the adjusted R^2 is an abysmal 0.0775 for 7.75% of the variance explained by the model. Notably, all coefficients are negative, indicating a one-unit positive change in their value decrease the STRESS variable. This aligns with common sense as an increase in family cohesion, self-esteem, grades, and attachment to their school all indicate a positive mental outlook / state of life, whereas a lack of any of these may be due to stressful events in their life.

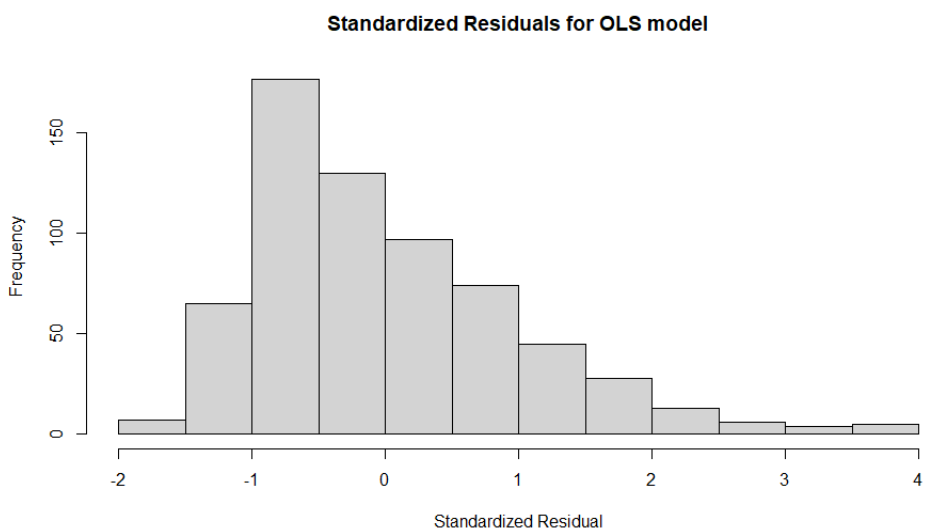

Figure 3. Histogram of standardized residuals for the OLS model. This shows a non-normal curve, with a positive skew based on the tail of data towards the maximum value of 9.

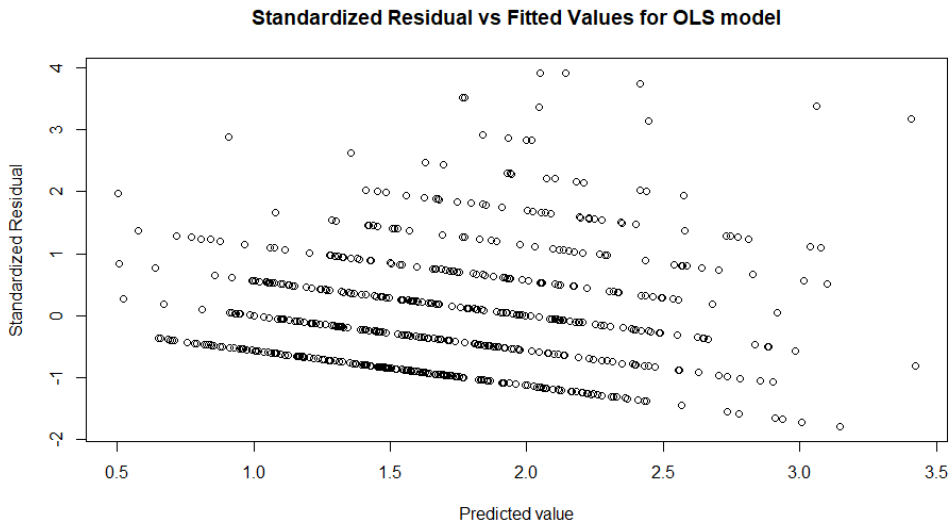

Figure 4. Scatterplot of standardized residuals versus the predicted STRESSS values for the OLS model. There is little evidence of heteroscedasticity, with about 30 points above the standard |2| cutoff (4.6%).

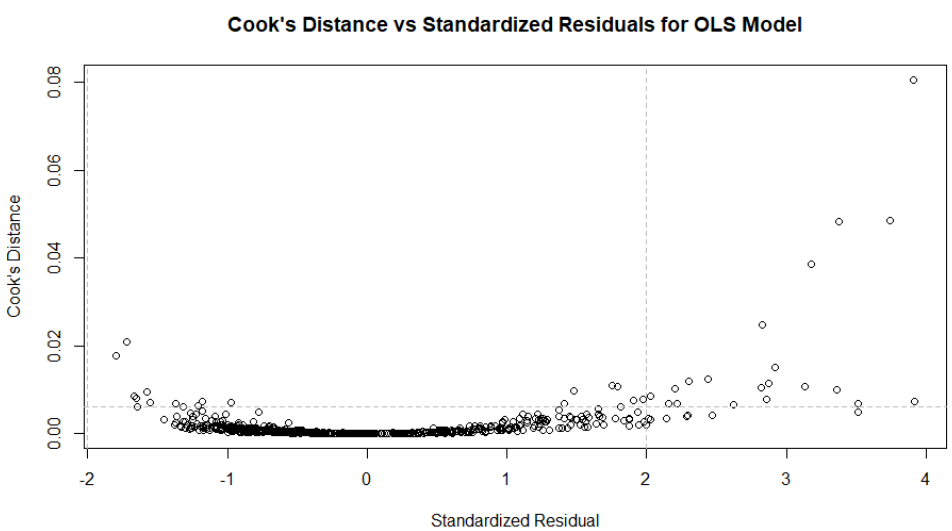

Figure 5. Scatterplot of Cook's Distance versus the standardized residuals for the OLS model. There are again a handful of points to be investigated both by a Cook's distance cutoff of 0.00614 (horizontal line), and a standardized residual cutoff of 2. The top 5 points there are all between about 0.025 and 0.08 on the Cook's distance axis as well as above 2.75 for the standardized residual, which would persuade us to follow up on those points to see whether they are anomalies and should be removed from the dataset.

a. While the latter two diagnostic plots are particularly impressive, the histogram of standardized residuals leaves room for improvement where it looks closer to the Poisson distribution as we will see later. The most indicative part of the lack of goodness-of-fit for this multiple linear model is the adjusted R^2 which may be the lowest of any we've seen so far in this class.

b. Also, as there are values for STRESS listed above the intercept estimate of 5.7, and all of the explanatory variables cannot go lower than 0, there is no way for the model to go as high as 6, 7, 8, or 9, of which there are about 30 observations. These are potentially related to the high standardized residuals. After pulling the std.res > 2 set of datapoints, all but one are a STRESS value between 6 and 9, indicating that these are the datapoints "messing" with the model, which makes sense if the model can only get close to the value if the observation reports a 0 for all explanatory variables.

3. Create a transformed variable on Y that is LN(Y). Fit an OLS regression model to predict LN(Y) using COHES, ESTEEM, GRADES, SATTACH as explanatory variables (X). Obtain the typical diagnostic information and graphs. Discuss how well this model fits. Obtain predicted values (LN(Y)_hat) and plot them in a histogram. What issues do you see? Does this correct the issue?
   a. Firstly, we can solve the issue of finding undefined values for ln(0) by adding a small amount to each STRESS value, here using 1e-11 so the model can be estimated.

```
> summary(ols_ln)

Call:
lm(formula = log(STRESS) ~ COHES + ESTEEM + GRADES + SATTACH,
    data = mdcl)

Residuals:
    Min      1Q  Median      3Q     Max
-22.826 -15.255   6.990   9.414  14.761

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.46966    3.98895   2.123   0.0341 *
COHES       -0.11878    0.04825  -2.462   0.0141 *
ESTEEM      -0.11619    0.13269  -0.876   0.3815
GRADES      -0.15425    0.16143  -0.955   0.3397
SATTACH     -0.15872    0.09694  -1.637   0.1021
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.19 on 646 degrees of freedom
Multiple R-squared:  0.03648,	Adjusted R-squared:  0.03052
F-statistic: 6.115 on 4 and 646 DF,  p-value: 7.82e-05
```

Output 2. Summary output for the ln(STRESS) OLS model. The adjusted R^2 is worse than the original model, and only the intercept and COHES are significant, both to only the 0.05 threshold. It is likely the rest of the diagnostic plots will show this model underperforming the original model. There is approximately a 38% and 34% chance of the ESTEEM and GRADES variables, respectively, having a true coefficient of 0 for this model. Notably, all coefficients are negative, indicating a one-unit positive change in their value decrease the STRESS variable. This aligns with common sense as an increase in family cohesion, self-esteem, grades, and attachment to their school all indicate a positive mental outlook / state of life, whereas a lack of any of these may be due to stressful events in their life.

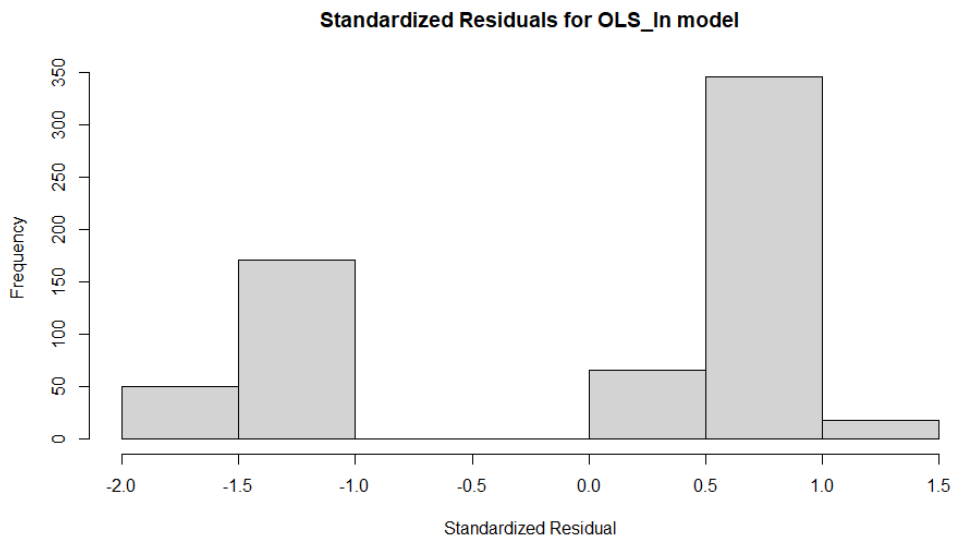

Figure 6. Histogram of standardized residuals for OLS_ln (natural log) model. The result matches the lack of accuracy for the model, with no standardized residuals between -1 and 0. This plot does not exhibit a normal curve, unless it is treated as bimodal, which it should not be.

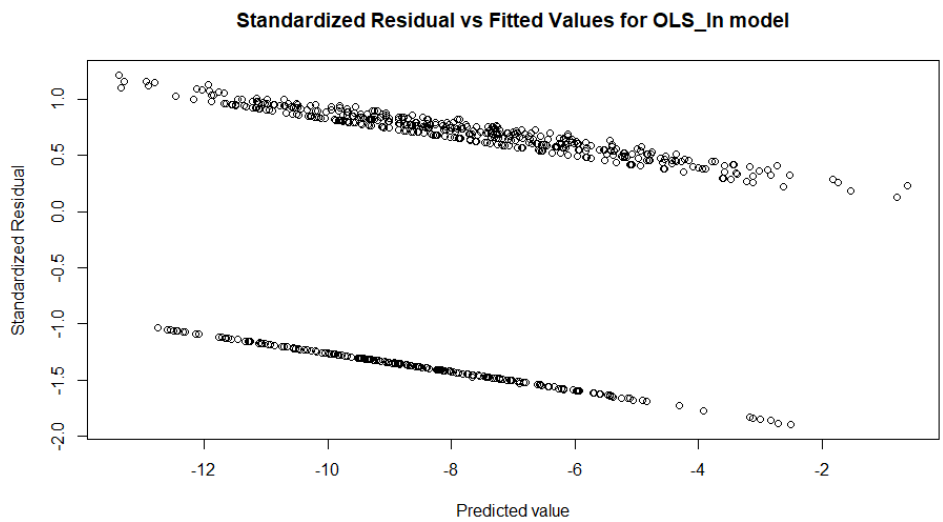

Figure 7. Scatterplot of standardized residuals versus the predicted value based on the natural log transformation of the STRESS variable. Notably, they are predicting all negative values due to the ln(1e-11) being -25.3, and with the high count of those the whole model is pulled downwards. There is also a clear delineation between two linear components of dataset.

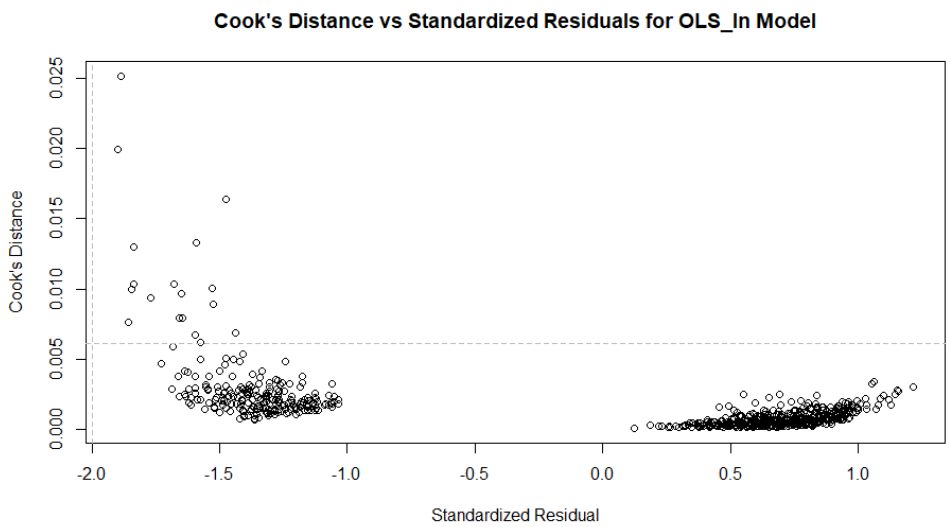

Figure 8. Scatterplot of Cook's distance versus standardized residuals for the OLS_ln model. Once again the model does not perform well, while it stays within the |2| boundaries of standard deviations, the Cook's distance (horizontal line) shows about 20points above the threshold to be investigated. Outside of the lack of observations with a standardized residual between -1 and 0, the lessening of high leverage points could point towards improvement if the rest of the statistics were as good as that singular statistic.

b. Overall this model is not an improvement on the first model and should be discarded in favor of the non-ln() model. Ideally the Poisson or ZIP model will improve upon these results.
c. It seems to have caught the higher ranging values more appropriately, with all 28 previous points of issue being contained below a Cook's distance of 0.003 and between std.res 0.2 and 1.2. While it is accounting for the higher values better, 48/50 of the top Cook's distance values are for STRESS values of 0 (in our model 1e-11). This model attempts to gather in all of the STRESS values better than the original, but in doing so becomes less able to account for STRESS values of 0.

4. Use the glm() function to fit a Poisson Regression for STRESS (Y) using COHES, ESTEEM, GRADES, SATTACH as explanatory variables (X). Interpret the model's coefficients and discuss how this model's results compare to your answer for part 3). Similarly, fit an over-dispersed Poisson regression model using the same set of variables. How do these models compare?

```
> summary(pm <- glm(STRESS ~ COHES + ESTEEM + GRADES + SATTACH, family = "poisson", data = md))

Call:
glm(formula = STRESS ~ COHES + ESTEEM + GRADES + SATTACH, family = "poisson",
    data = md)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.7111  -1.5989  -0.2914   0.7107   3.6424

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.734513   0.234066  11.683  < 2e-16 ***
COHES       -0.012918   0.002893  -4.466 7.98e-06 ***
ESTEEM      -0.023692   0.008039  -2.947  0.00321 **
GRADES      -0.023471   0.009865  -2.379  0.01735 *
SATTACH     -0.016481   0.005783  -2.850  0.00437 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1349.8  on 650  degrees of freedom
Residual deviance: 1245.4  on 646  degrees of freedom
AIC: 2417.2

Number of Fisher Scoring iterations: 5
```

Output 3. Summary output for the Poisson model. The variables listed are all significant at least at the 0.05 threshold which is an improvement on the other models thus far. Running a X^2 z-test for the difference in deviance and 4 degrees of freedom, we get a value of 0, indicating we can reject the null hypothesis that the intercept-only model is a better model than this model. Said another way, we can confirm at least one of these variables is significant in the true model and has a non-zero coefficient.

a. These variable estimates are related to the log-linked function, so they are representative of the effect of a one-unit change in the variable on the log-contained function that creates the STRESS variable. Notably, they are all negative, indicating a reduction in the value that eventually is passed to the exponent function. An example is holding all else equal, a one-unit increase in COHES correlates to a 1.283% decrease in the number of STRESS events which starts at 15.402 with all 0 entries for the explanatory variables through exponentiation. Similarly, a one-unit increase in ESTEEM, GRADES, and SATTACH lead to a 2.341%, 2.320%, and 1.635% decrease in STRESS respectively.

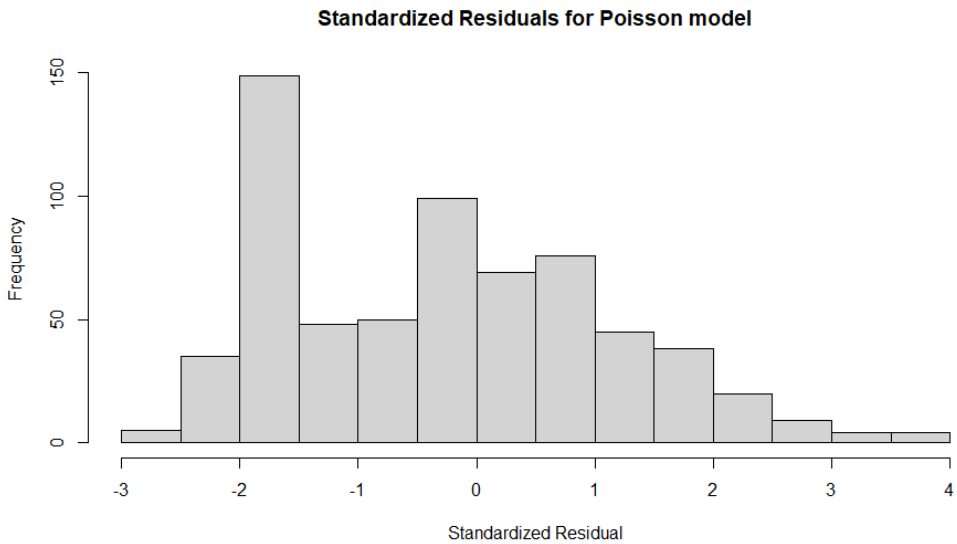

Figure 9. Histogram of standardized residuals for the Poisson model. This shows a relatively normal curve with a spike around -2, likely due to the preponderance of 0s in the dataset. As noted previously, there are 221 0s out of the total 651 observations. The 0 observations make up 222/223 of the lowest standardized residuals, which tracks as they are the lower values. The fact that this model's standardized residuals at all resemble a normal distribution appears to be a step up from previous models.

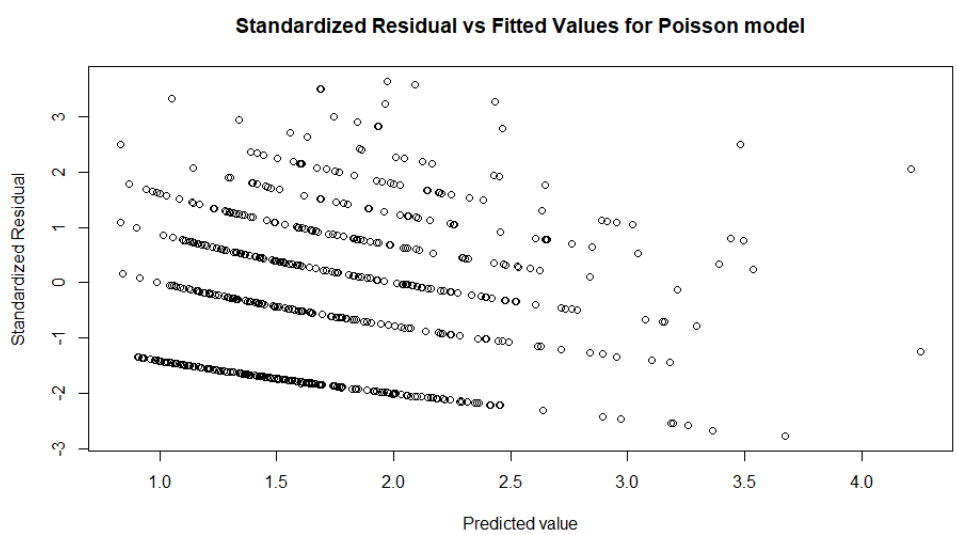

Figure 10. Scatterplot of standardized residuals versus fitted values for the Poisson model. There appears to be a homoscedastic spread, with many observations (about 80) with standardized residual past our typical |2| threshold.

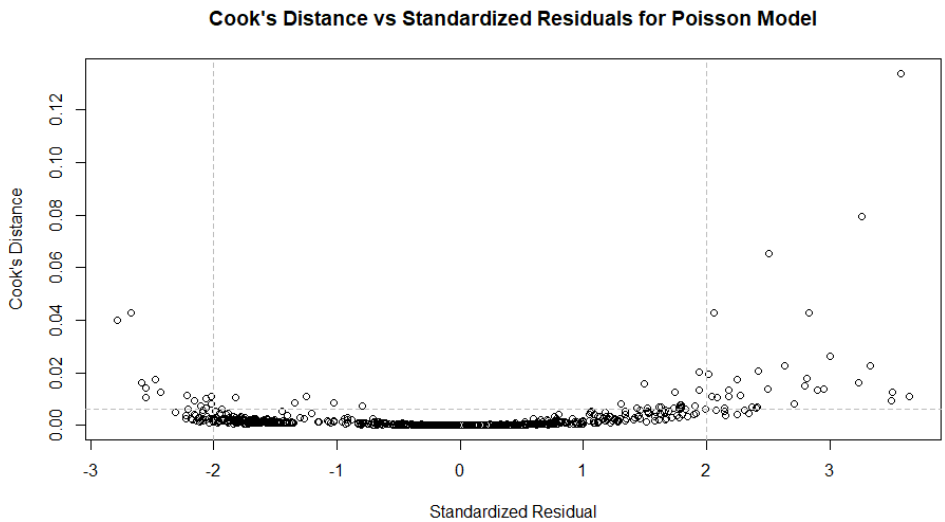

Figure 11. Scatterplot of Cook's distance versus standardized residuals for the Poisson Model. This shows a typical pattern as we've seen in previous assignments, with some outliers above the Cook's distance cutoff of 4/651 ≈ 0.006 and an absolute standardized residual above 2, with at least 7 points that are most noteworthy and can be investigated to understand why they are so far off in this model.

b. I am understanding the "overdispersed Poisson" to be the Negative Binomial model and will operate as such moving forward for this assignment.

```
> summary(nb)

Call:
glm.nb(formula = STRESS ~ COHES + ESTEEM + GRADES + SATTACH,
    data = md, link = log, init.theta = 1.865329467)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0179  -1.3900  -0.2214   0.4882   2.3199

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.759032   0.341531   8.078 6.56e-16 ***
COHES       -0.013391   0.004136  -3.238  0.00121 **
ESTEEM      -0.023058   0.011477  -2.009  0.04453 *
GRADES      -0.024360   0.013969  -1.744  0.08118 .
SATTACH     -0.016750   0.008296  -2.019  0.04349 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(1.8653) family taken to be 1)

    Null deviance: 792.47  on 650  degrees of freedom
Residual deviance: 738.53  on 646  degrees of freedom
AIC: 2283.6

Number of Fisher Scoring iterations: 1


              Theta:  1.865
          Std. Err.:  0.257

 2 x log-likelihood:  -2271.590
```

Output 4. Summary output for Negative Binomial model based on all given explanatory variables (outside of NEWID). The statistical significance was lost a bit moving to this model, dropping the all-0.05

threshold with GRADES at a p-value of 0.0812. This model predicts similar values to the Poisson Model, starting with 15.785 but with percent decrease per unit change of COHES, ESTEEM, GRADES, and SATTACH of 1.33%, 2.28%, 2.41%, and 1.66% respectively. In other words, there is a slight increase in the effect of COHES, decrease in ESTEEM, increase in GRADES, and increase in SATTACH, though not significantly.

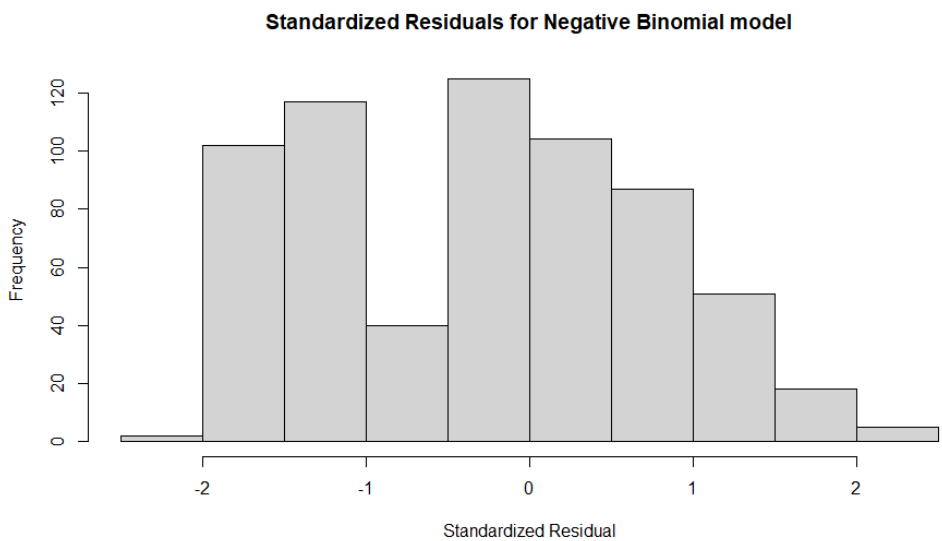

Figure 12. Histogram of standardized residuals for the Negative Binomial model. There appears to be a relatively normal distribution with a slight dip from -0.5 to -1, creating a bit of a bimodal pattern.

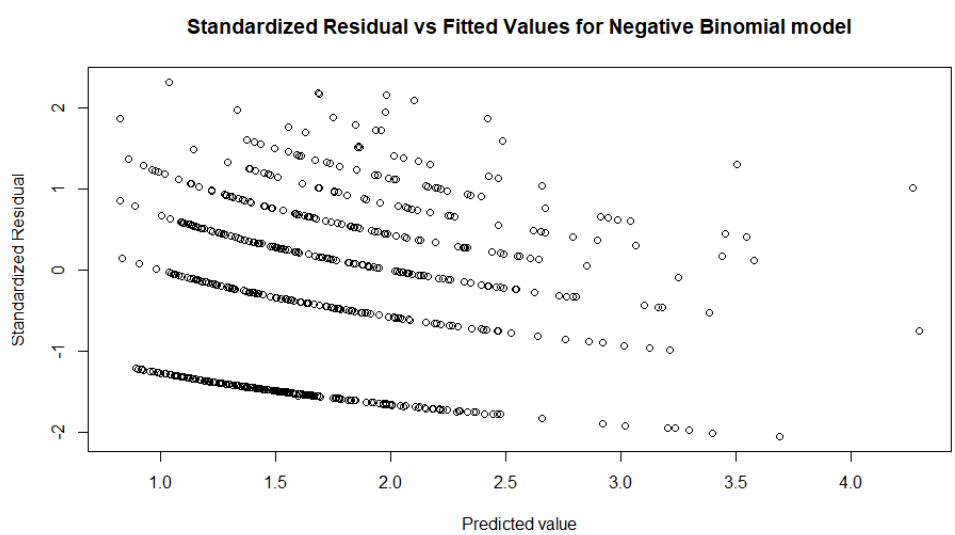

Figure 13. Scatterplot of standardized residuals vs fitted values for the Negative Binomial model. Here there is little evidence of heteroscedasticity, and only 5 observations with an absolute standard deviation above 2. This is the best percentage of all the models so far for coverage within the -2 to 2 range.

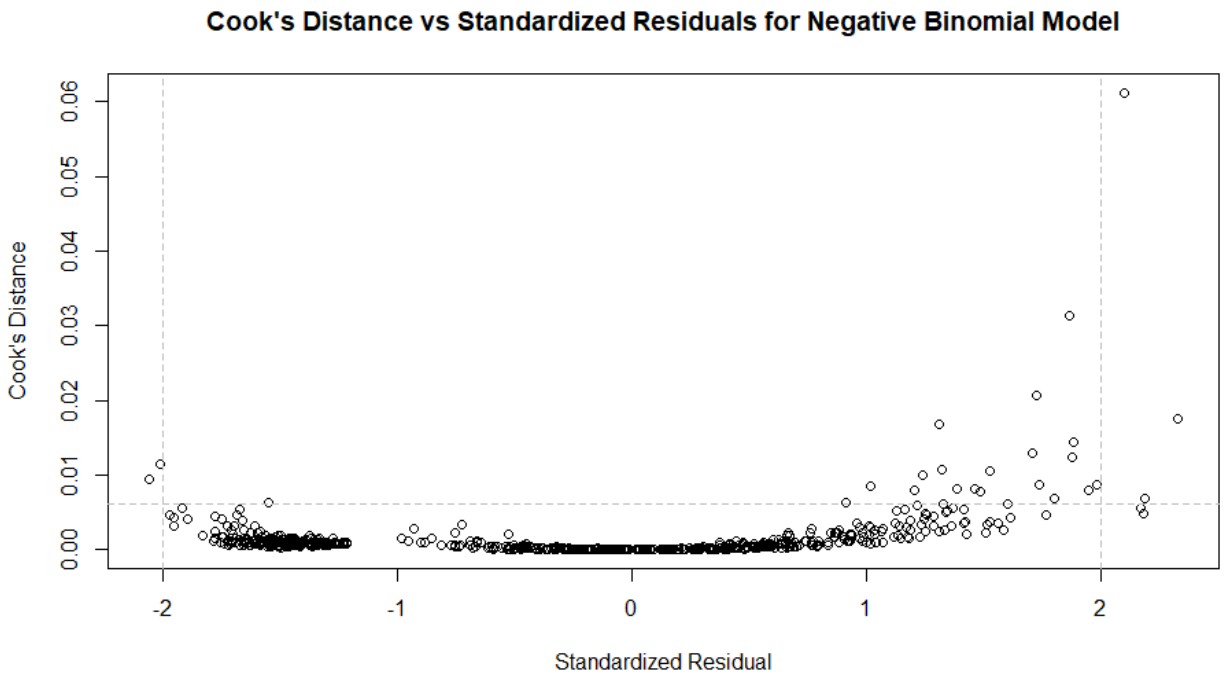

Figure 14. Scatterplot of Cook's distance versus standardized residual for the Negative Binomial model. While there are far less observations outside of our "comfort zones" (within abs.std.res <2 and Cook's distance less than ~0.006, there are still a handful of points that could be investigated to see if they should be removed for some reason.

5. Based on the Poisson model in part 4), compute the predicted count of STRESS for those whose levels of family cohesion are less than one standard deviation below the mean (call this the low group), between one standard deviation below and one standard deviation above the mean (call this the middle group), and more than one standard deviation above the mean (high). What is the expected percent difference in the number of stressful events for those at high and low levels of family cohesion?
    a. The mean of the low COHES group STRESS Poisson prediction is 2.503, and the mean of the high group prediction is 1.888, indicating a 210.624% increase in STRESS events prediction when moving from a high COHES group to the low COHES group. In other words, a student with a low family cohesion is likely to have twice the stress events as a similar student with high family cohesion. This tracks with common sense where one category of these stress events is family related (abuse, parent divorce, etc.) and that would likely align with a lessening of the cohesion for the family unit.

6. Compute the AICs and BICs from the Poisson Regression and the over-dispersed Poisson regression models from part 4). Is one better than the other?
    a. In terms of AIC and BIC, lower is better for a lower amount of prediction error. In our models, the AIC for the Poisson and Negative Binomial models were 2417.219 and

2283.59, respectively, and BIC of 2439.612 and 2310.461, also respectively. The Negative Binomial distribution performs better with a lower value in both AIC and BIC. This tracks with the note in 1a where the distribution better aligns with a typical NB distribution as opposed to Poisson.

b. For those reasons, we would likely opt to use the NB model if choosing between it and the Poisson model as a final model.

7. Using the Poisson regression model from part 4), plot the deviance residuals by the predicted values. Discuss what this plot indicates about the regression model.

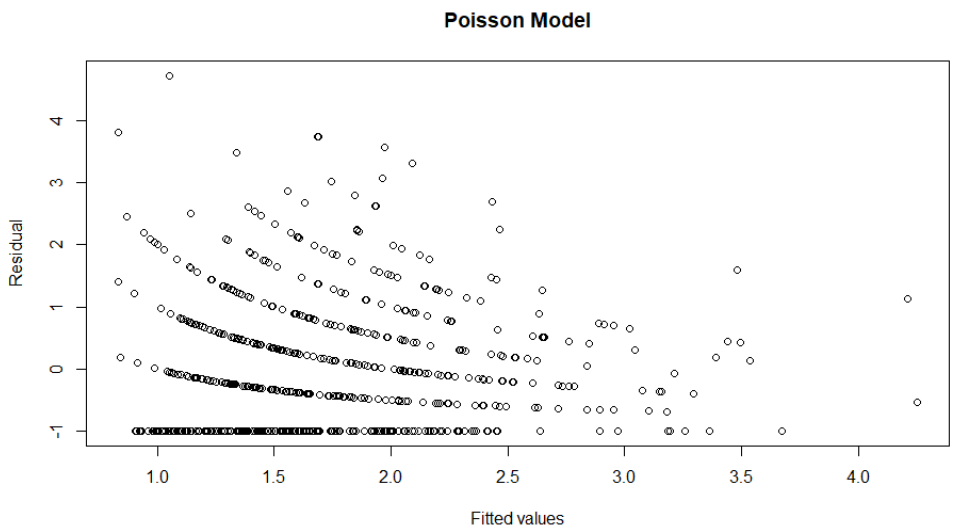

Figure 15. Scatterplot of residual versus fitted values for the Poisson model on STRESS.

a. The wave pattern is not unusual for graphing counts and predicted outcomes. More importantly we want to look for the number of points past 3 either positive or negative. Here we have about 10 points past the 3 cutoff, which is not terrible for the 651 observations. Overall this model performs well according to this perspective. When compared to other models, we would hope the values would all be kept in the -3 to 3 thresholds, but overall this model seems to perform well in its current state.

8. Create a new indicator variable (Y_IND) of STRESS that takes on a value of 0 if STRESS=0 and 1 if STRESS>0. This variable essentially measures is stress present, yes or no. Fit a logistic regression model to predict Y_IND using the variables using COHES, ESTEEM, GRADES, SATTACH as explanatory variables (X). Report the model, interpret the coefficients, obtain statistical information on goodness of fit, and discuss how well this model fits. Should you rerun the logistic regression analysis? If so, what should you do next?

```
> summary(log_model)

Call:
glm(formula = y_ind ~ COHES + ESTEEM + GRADES + SATTACH, family = binomial(link = "logit"),
    data = md)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.9069  -1.3283   0.7829   0.9366   1.2693

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.516735   0.737131   4.771 1.83e-06 ***
COHES       -0.020733   0.008751  -2.369   0.0178 *
ESTEEM      -0.018867   0.023741  -0.795   0.4268
GRADES      -0.025492   0.028701  -0.888   0.3744
SATTACH     -0.027730   0.017525  -1.582   0.1136
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 834.18  on 650  degrees of freedom
Residual deviance: 811.79  on 646  degrees of freedom
AIC: 821.79

Number of Fisher Scoring iterations: 4
```

Output 5. Summary output for the logistic model predicting Y_ind from COHES, ESTEEM, GRADES, and SATTACH. Only COHES is significant at any level, where ESTEEM is nearly 50% likely to be a nonsignificant variable with a coefficient of 0.

a. The model uses a printout of log-odds, so these are related to the log changes in the odds ratio. The lack of significance in the coefficients could point to the idea that the chance of a stressful event happening in the first place is more related to the world outside of the student than the students creating that stressful situation themselves. Additionally, we are treating the jump from 0 to 1 as more significant than a jump from 1 to 2 and so on, which for stressful life events may not be as true as how long a shower takes in a day. To explain what the coefficients are here, we exponentiate the coefficients to get their “real” values. For example, the coefficient gives us e^-.0207 = 0.97951, so a 2.049% drop in having STRESS > 0 as COHES increases. The same negative value indicates the likelihood of having any STRESS events drops as these variables increase, with 1.869% drop for a one unit increase in ESTEEM, 2.517% drop for a unit increase in GRADES, and 2.735% drop for a unit increase in SATTACH.
b. That being said, it’s noted in the caption of Output 5 that most of these values are not significant so the model needs some work before using it to make predictions. Likely we would look to first iterate over the predictor variables to ensure we have the best set, if not look for other variables that could be added to the dataset like what grade level they’re in to better predict the likelihood of stress events.

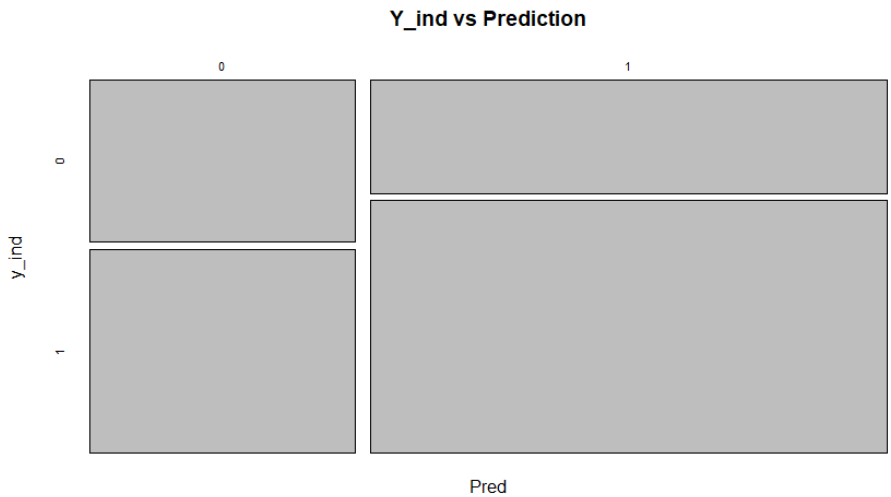

Figure 16. Mosaic plot of the prediction versus Y-ind. This shows 60.676 (395/651) of the observations were correctly predicted, which is not a great percentage though better than a random 0/1 guess of 50/50. Moreover, there were 297 true positives (Y_ind = 1 and Pred = 1), followed by 133 false negatives (1/0), 123 false positives (0/1), and finally 98 true negatives (0/0).

9. It may be that there are two (or more) process at work that are overlapped and generating the distributions of STRESS(Y). What do you think those processes might be? To conduct a ZIP regression model by hand, fit a Logistic Regression model to predict if stress is present (Y_IND), and then use a Poisson Regression model to predict the number of stressful events (STRESS) conditioning on stress being present. Is it reasonable to use such a model? Combine the two fitted model to predict STRESS (Y). Obtained predicted values and residuals. How well does this model fit? HINT: You have to be thoughtful about this. It is not as straight forward as plug and chug!
    a. I think it's not unreasonable to examine whether there a Zero-Inflated model is appropriate here due to the high percentage of 0 entries present in the data. These stress events are typically rare enough, with unfortunate students having one or multiple. For the ZIP / ZINB regression, the two underlying processes would be the model predicting if there was at least one stress event, and if so then how many were there. The first portion is the Zero-Inflated addition, whereas the second part involves the Poisson or Negative Binomial. While this may not be as good of a situation to apply this thought process compared to the shower length example but should still work alright for our training purposes.

```
> summary(cp)

Call:
glm(formula = STRESS ~ COHES + ESTEEM + GRADES + SATTACH, family = "poisson",
    data = md_combined[md_combined$y_ind == 1, ])

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.6875  -0.8596  -0.2063   0.5268   2.9480

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.311700   0.239989   9.633   <2e-16 ***
COHES       -0.006254   0.002945  -2.124   0.0337 *
ESTEEM      -0.019521   0.008107  -2.408   0.0160 *
GRADES      -0.014661   0.009689  -1.513   0.1302
SATTACH     -0.008002   0.005836  -1.371   0.1703
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 415.81  on 429  degrees of freedom
Residual deviance: 380.26  on 425  degrees of freedom
AIC: 1552.1

Number of Fisher Scoring iterations: 5
```

Output 6. Summary call for Poisson model run on the y_ind = 1 observations. While there are only two non-intercept values that are significant to the 0.05 threshold, the general trend with negative coefficients remains consistent.

b. Running the X^2 for the difference in deviances, we get a value of 3.58e-7, so we can reject the null and confirm that this model still works better than the intercept-only model for these non-zero STRESS observations.
c. For those observations, we can see the following standard diagnostic plots, given that we’ve already confirmed the model assigned a 0 prediction for approximately 1/3 of the datapoints.

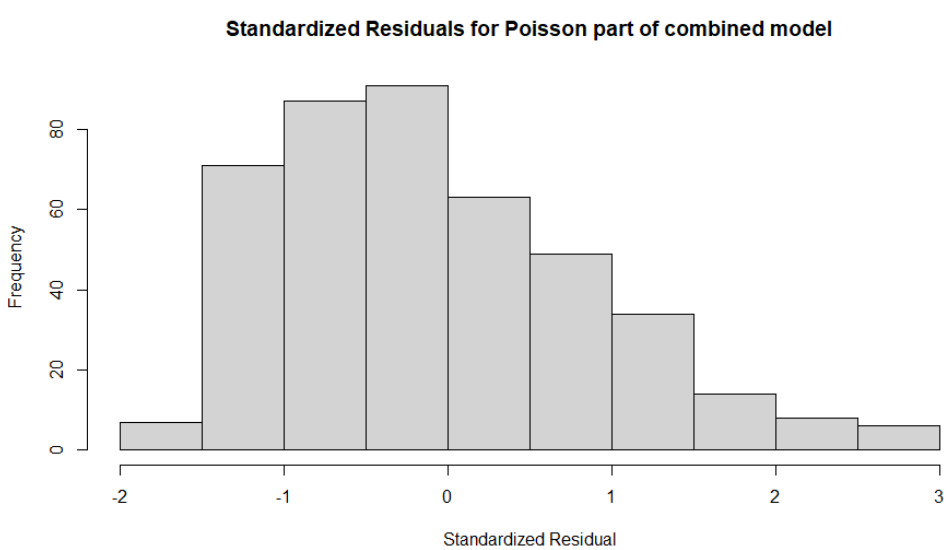

Figure 17. Histogram of standardized residuals for the Poisson model’s predictions on the y_ind = 1

observations. This shows an expected Poisson distribution, with a few residuals as high as 3 and as low as -2, which is fairly good for a model and has no huge outliers based on this distribution.

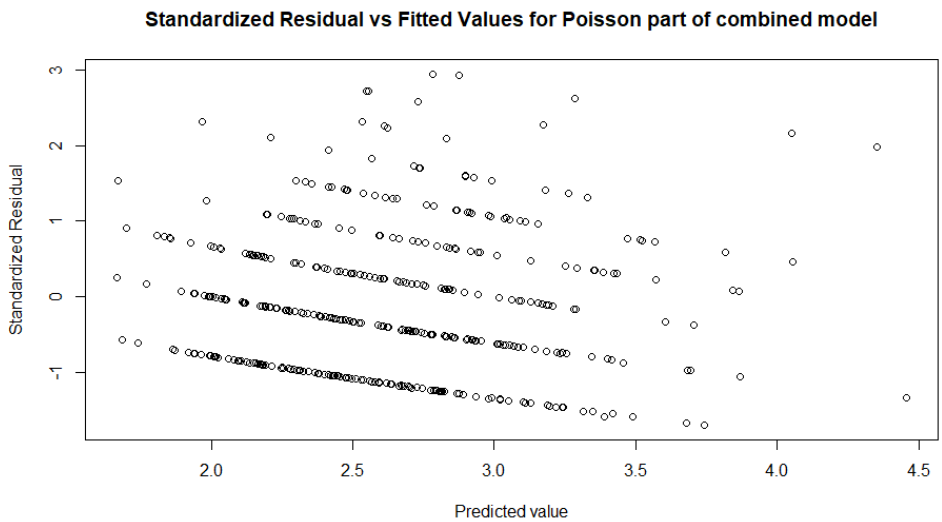

Figure 18. Scatterplot of standardized residuals versus predicted value for the Poisson portion of the combined model where y_ind = 1. There is little evidence of heteroscedasticity and the apparent lines are typical in count vs predicted graphs. As indicated in the Figure 17 caption, the standardized residuals remain below 3, so the model is appropriate if the goodness-of-fit is based on this graph alone.

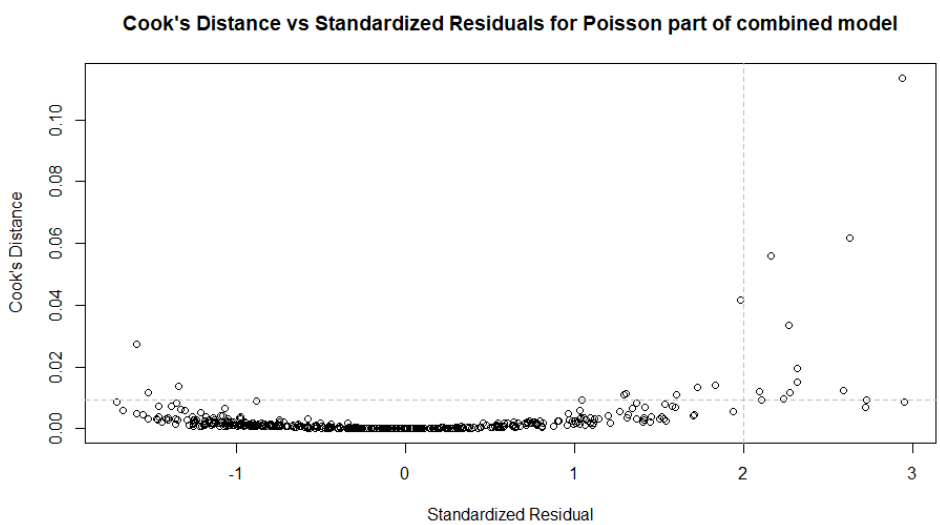

Figure 19. Scatterplot of Cook's distance versus standardized residual for the Poisson portion of the combined model. There are a few points in the "danger zone" of being above the Cook's cutoff as well as >2 for the standardized residual cutoff. These points should be investigated in case they merit removal for some outside force impacting the STRESS value for these students who have at least one stress event.

d. While this model performs admirably, it was noted in the summary output that the statistical significance of our variables indicates the model cannot be trusted in its

predictions for STRESS. While it appears to function well enough based on the diagnostic plots, more work is needed to ensure the model fully operates as well as possible. As noted above, it's worth considering adding variables if more can be collected or consider another prediction model for the non-zero entries like Negative Binomial.

10. Use the pscl package and the zeroinfl() function to Fit a ZIP model to predict STRESS(Y). You should do this twice, first using the same predictor variable for both parts of the ZIP model. Second, finding the best fitting model. Report the results and goodness of fit measures. Synthesize your findings across all of these models, to reflect on what you think would be a good modeling approach for this data.

```
> summary(zip)

Call:
zeroinfl(formula = STRESS ~ COHES + ESTEEM + GRADES + SATTACH, data = md_zip, dist = "poisson")

Pearson residuals:
    Min      1Q  Median      3Q     Max 
-1.4534 -0.9136 -0.2166  0.6257  3.9954 

Count model coefficients (poisson with log link):
             Estimate Std. Error z value Pr(>|z|)    
(Intercept)  2.641693   0.272349   9.700  < 2e-16 ***
COHES       -0.008258   0.003416  -2.418  0.01561 *  
ESTEEM      -0.026068   0.009206  -2.832  0.00463 ** 
GRADES      -0.019553   0.010914  -1.792  0.07320 .  
SATTACH     -0.010485   0.006673  -1.571  0.11611    

Zero-inflation model coefficients (binomial with logit link):
             Estimate Std. Error z value Pr(>|z|)   
(Intercept) -2.835428   0.983250  -2.884  0.00393 **
COHES        0.018917   0.012124   1.560  0.11869   
ESTEEM      -0.004328   0.032777  -0.132  0.89496   
GRADES       0.014330   0.037731   0.380  0.70410   
SATTACH      0.024838   0.024083   1.031  0.30238   
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 16 
Log-likelihood: -1134 on 10 Df
```

Output 7. Summary output for the ZIP model. Notably none of the variables listed are statistically significant in predicting the jump from 0 to any STRESS, and the statistical significance of the explanatory variables is not strong within the count predictions as it was during the Negative Binomial model application.

a. The method I went for next is to run a ZIP regression on each subset of variables, with 3, 2, or 1 of the explanatory variables, comparing them with a difference of log-likelihoods using logLik and comparing to the X^2 distribution, as well as the AIC and BIC, giving the results below:

Model #	Variables	AIC	BIC	Log-Likelihood	Df	X^2
1	COHES, ESTEEM, GRADES, SATTACH	**2288.802**	2333.587	-1134.401		

2	COHES, ESTEEM, GRADES	**2289.745**	2325.573	-1136.873	1	0.1159302
3	COHES, ESTEEM, SATTACH	**2288.734**	2324.562	-1136.367	1	0.1608674
4	COHES, GRADES, SATTACH	2293.358	2329.186	-1138.679	1	0.03861113
5	ESTEEM, GRADES, SATTACH	2296.664	2332.492	-1140.332	1	0.01487591
6	COHES, ESTEEM	2291.855	**2318.726**	-1139.927	2	0.06309106
7	COHES, SATTACH	2294.998	**2321.869**	-1141.499	2	0.02875443
8	GRADES, SATTACH	2308.725	2335.596	-1148.362	2	**0.0009297478**
9	COHES, GRADES	2296.793	2323.664	-1142.397	2	0.01835623
10	ESTEEM, SATTACH	2298.349	2325.22	-1143.175	2	0.01244036
11	ESTEEM, GRADES	2302.203	2329.074	-1145.102	2	0.004746678
12	COHES	2302.268	**2320.182**	-1147.134	3	0.005251302
13	ESTEEM	2308.348	2326.262	-1150.174	3	0.001262134
14	GRADES	2323.496	2341.41	-1157.748	3	**0.00003418696**
15	SATTACH	2314.956	2332.87	-1153.478	3	**0.0002635868**

Table 1. List of models, variables, AIC, BIC, log-Likelihood, degrees of freedom and X^2 based on the difference in log-Likelihood versus Model 1. Additionally, the top three values for the relevant columns have been bolded.

b. As seen on the table, different statistics provide different answers as to which models would be best here. AIC favors the bigger models, BIC a bit of a mix, and X^2 the smaller models. In an attempt to narrow them down, we will look at those with <0.05 for X^2 then choose the best based on the AIC/BIC combo. For that, we'll have 4, 7, 8, 9, 10, 11, 12, 13, 14, and 15. From them, multiplying the AIC and BIC to get the minimal among that combination is model 7 with a combined 5,328,685.

```
> summary(zip_7 <- zeroinfl(STRESS ~ COHES + SATTACH, data = md_zip, dist = "poisson"))

Call:
zeroinfl(formula = STRESS ~ COHES + SATTACH, data = md_zip, dist = "poisson")

Pearson residuals:
    Min      1Q  Median      3Q     Max 
-1.3921 -0.9101 -0.2409  0.5914  4.0319 

Count model coefficients (poisson with log link):
             Estimate Std. Error z value Pr(>|z|)    
(Intercept)  1.905773   0.175578  10.854  < 2e-16 ***
COHES       -0.012241   0.003235  -3.784 0.000154 ***
SATTACH     -0.016467   0.006481  -2.541 0.011064 *  

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value Pr(>|z|)    
(Intercept) -2.87292    0.67606  -4.249 2.14e-05 ***
COHES        0.01934    0.01114   1.737   0.0824 .  
SATTACH      0.02917    0.02299   1.269   0.2045    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 12
Log-likelihood: -1141 on 6 Df
```

Output 8. Summary output for our Model 7 using COHES and SATTACH. While the Zero-inflation model coefficients are mostly not statistically significant, the Count model coefficients seem useful per the 0.05 threshold. These values give us a fair idea of what might be useful for the overall model with the variables provided in the dataset.

- As a final note, it seems that the models work well enough but depending on the intent of the model's usage some work well than others. Between the Poisson and Negative Binomial models, the NB model performed better according to the diagnostic plots. For that reason and the distribution of the model, it appears Negative Binomial would be a fine model to use.
- Next looking at the Zero-inflated versions, while most performed well, the manual version we created appeared to predict worse than the zeroinfl() calls afterwards, with the improved version in our set being the model using COHES and SATTACH. For that reason, we could use this model if we intend to use a Zero-inflated model to predict, we could look at the difference between Poisson and Negative Binomial for the distribution call in zeroinfl(). After a quick check for the NB side of this function, it appears the same trends are true of the AIC, BIC, and X^2 though slightly improved compared to the Zero-inflated Poisson set.
- Further versions would include looking at a stepwise (or forward/backwards) approach to creating this model or adding interaction terms in case there are other relationships between the explanatory variables.